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2 DOCTORAL THESIS

3

4 **The study of Higgs properties in the**
5 **four-lepton final state at CMS**

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153 *Dedicated to my family who endured missed birthdays,*
154 *missed festivals and countless days alone so that I may*
155 *make history*

Chapter 1

Probing four lepton final-state for Higgs Cross section measurement

1.1 Introduction

Higgs particle has been the first fundamental scalar particle which was observed at LHC, CERN. Some years ago in 2012, both CMS and ATLAS collaborations announced its discovery [1, 2] with its mass around 125 GeV. Later on, these experiments have also studied its comprehensive properties [3, 4, 5, 6] which are found to be consistent with the SM Higgs.

In this chapter, measurements of fiducial cross sections (inclusive and differential) of Higgs in the $H \rightarrow 4\ell$ ($\ell = e, \mu$) decay channel using full Run 2 data recorded with CMS from collisions of proton-proton at $\sqrt{s} = 13\text{TeV}$ are presented which give a direct test of the perturbative QCD calculations of SM (Higgs sector) at TeV energy scale. Any deviation of the integrated or differential distributions from SM expectations would provide a direct hint to BSM contributions. These measurements with earlier or partial Run2 CMS data has also been performed Ref. [Khachatryan:2015yvw], [CMS:2015hja], [CMS:2016rqf], [Sirunyan:2017exp], [CMS:2018hhg]. Nearly similar measurements have already been reported by ATLAS experiment in $H \rightarrow \gamma\gamma$ [atlas-conf-2019-029], [7, 8], $H \rightarrow 4\ell$ [atlas-conf-2019-025], [9, 10] decay channels, combination of both channels [atlas-conf-2019-032], [11, 12] and by CMS experiment in $H \rightarrow \gamma\gamma$ decay channel [13].

The integrated fiducial cross section measurements are performed using collisions data of CMS (LHC) corresponding to 137.1 fb^{-1} integrated luminosity at $\sqrt{s} = 13\text{ TeV}$. It is important to minimize results' dependence on underlying models of Higgs production mechanism and its properties. Hence, cross sections are

181 extracted in the fiducial phase space that closely matches with the detector geom-
 182 etry as shown in Figure 1.1. The remaining model dependence is extracted by
 183 repeating the measurements for SM and other exotic models and reported as a
 184 distinct systematic uncertainty. All measurements includes the corrections for the
 185 finite detection efficiency and resolution effects. As $\sigma \times BR$ strongly depends on
 186 the mass of Higgs, measurements are extracted assuming $m_H = 125.09$ GeV, as
 187 measured from $H \rightarrow 2\gamma$ and $H \rightarrow 4\ell$ decay modes at CMS experiment of LHC
 [14].

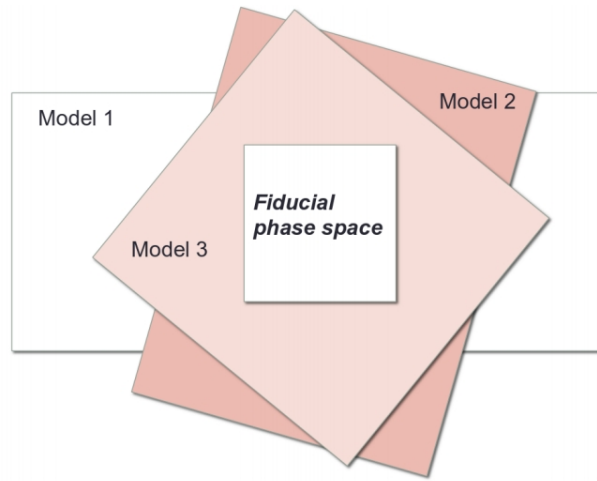


FIGURE 1.1: Fiducial phase definition among different models.

188
 189 The cross sections are also measured differentially as a function of kinematic
 190 observables. In this dissertation we study the differential observables which are
 191 sensitive to production mechanism of Higgs: transverse momentum and the ra-
 192 pidity of Higgs, the transverse momentum of leading jet and jet multiplicity.

193 The structure of this chapter follows as the datasets and MC or simulated sam-
 194 ples used are introduced in Section 1.1.1. The analysis strategy is described in Sec-
 195 tion 1.2, where event selection, fiducial phase space definition and strategy method
 196 for extracting inclusive and differential fiducial cross sections. In Section 1.3 details
 197 of systematic uncertainties affecting the cross section measurement are explained.
 198 In Section 1.4 results of measurements are compared with SM predictions.

1.1.1 Datasets, Simulation Samples and Theoretical Predictions

1.1.2 Datasets

Triggers and Datasets

The analysis uses full Run 2 (2016, 2017 and 2018) CMS data sample corresponding to 137.1 fb^{-1} integrated luminosity.

The datasets used are listed and documented in [CMS:2019chr]. The analysis relies on four different primary datasets (PDs), *DoubleMuon*, *MuEG*, *EGamma*, and *SingleMuon*, each of which combines a certain collections of HLT paths. To avoid duplicate events from different primary datasets, events are taken:

- from EGamma if they pass the diEle or triEle or singleElectron triggers,
- from DoubleMuon if they pass the diMuon or triMuon triggers and fail the diEle and triEle triggers,
- from MuEG if they pass the MuEle or MuDiEle or DiMuEle triggers and fail the diEle, triEle, singleElectron, diMuon and triMuon triggers,
- from SingleMuon if they pass the singleMuon trigger and fail all the above triggers.

The HLT paths used for 2018 collision data is listed in [CMS:2019chr].

Trigger Efficiency

Efficiency in data of combination of triggers used for the analysis is measured by considering 4ℓ events triggered by single lepton triggers. Among the reconstructed four leptons, is geometrically associated with an object that passes at least one single electron or muon triggers. Rest of three leptons are regarded as “probes”. In a 4ℓ event, the possible tag-probe pairs are 4 and are counted in denominator while computing efficiency. For each of three leptons which are probes, all matching trigger filter objects are collected. Then the matched trigger filter objects of the three probe leptons are combined in attempt to reconstruct any of the triggers used in the analysis. If any of the analysis triggers can be formed using the probe leptons, probes are call passing probes which are counted as numerator for the efficiency measurement.

This technique is used to measure efficiency on MC and the comparison with data is exploited to estimate the reliability of simulation. Fig. 1.2 shows the measured efficiency both in data and MC (2018) varied with minimum p_T of leptons. MC efficiency seems to describe data well within the uncertainties.

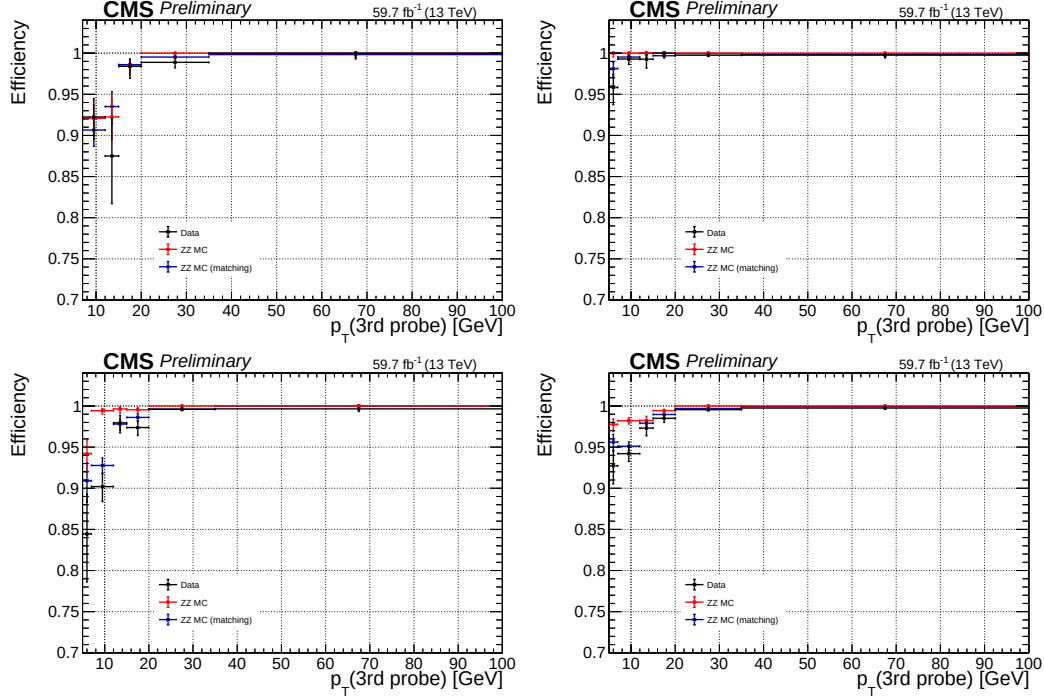


FIGURE 1.2: Measured trigger efficiency in 2018 data where the event collection is made using single electron and muon triggers: $4e$ (top left), 4μ (top right), $2e2\mu$ (bottom left) and 4ℓ (bottom right) final states.

1.1.3 Simulation Samples and Theoretical Predictions

Simulations of the interesting events help us build a realistic model of experiment in high energy. It enables us for a detailed study of physics processes (signal and background). Particularly for proton-proton collisions, generation of simulated events can be separated in many stages. For this analysis, in particular, descriptions of the production of SM Higgs through process of gluon-gluon fusion (Fig. 1.3) are acquired using POWHEG V2 [15, 16] generator. At accuracy level of next-to-next-to-leading order (i.e. NNLO) in perturbative QCD (pQCD), $gg \rightarrow H$ process description is obtained by partonic level MC generator NNLOPS.

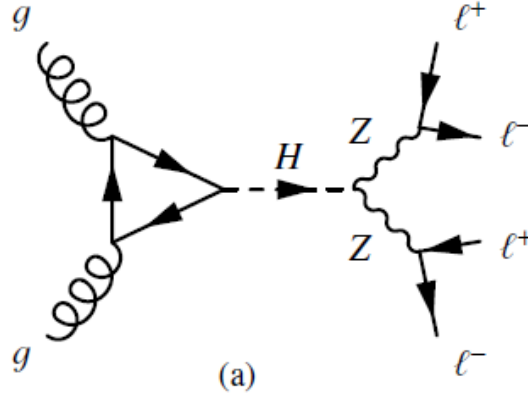
The POWHEG, provides perturbative QCD calculations at next-to-leading (NLO) accuracy with an connection to parton-shower programs. Using the POWHEG, Higgs production via $gg \rightarrow H$ with zero associated jets can be described at accuracy level of NLO. Finite masses of top and bottom quarks has been considered for $gg \rightarrow H$ production for all generators. The SM Higgs boson production by

Signal Samples

Descriptions of the SM Higgs boson production are obtained using the POWHEG V2 [17, 18, 19] generator for the five main production modes (ggH , VBF, WH, ZH and ttH) as described in section ?? [20], [21], [22], [23]. However, WH and ZH production modes are simulated by the MINLO HVJ extension of the POWHEG [23]. Higgs decay to four lepton is simulated by JHUGEN generator [24]. For the WH, ZH and ttH modes, Higgs is allowed to decay to $H \rightarrow ZZ \rightarrow 2\ell 2X$ such that two leptons from 4-lepton events are produced from the decay of associated W, Z bosons or top quarks are included in simulation. Showering of parton-level events is done using PYTHIA8.209, and in all cases matching is performed by allowing QCD emissions at all energies in the shower and vetoing them afterwards according to the POWHEG internal scale. All samples are generated with the NNPDF 3.1 NLO parton distribution functions (PDFs) [25]. The list of signal samples with the corresponding cross sections are shown in [CMS:2019chr]. These samples are also used to asses model dependence of measurement results are indicated in the Table ??.

The MC samples generated by POWHEG and PYTHIA8.209 are used to extract fiducial phase space acceptance and reconstruction efficiencies for individual models separately as explained in Section 1.2.4. These samples, with the sample generated by using the NNLOPS with alternate representation of $gg \rightarrow H$ process, are utilized for the comparison of measurement results with the theoretical predictions based on SM calculation in Section 1.4.

All SM production modes have been produced individually to approximate the dependence of measurements on underlying assumptions of Higgs production mechanism. The details of the method used to asses the model dependence will be described in Section 1.3. All these MC samples are produced using POWHEG for the representation of QCD effects at NLO in Higgs production, and by using

FIGURE 1.3: $gg \rightarrow H$ process fynman diagram.

273 JHUGEN [24, 26, 27] to represent such exotic resonances decays to four-leptons
 274 containing all the spin correlations.

275 Background Samples

276 For the estimations of the backgrounds using MC samples, the dominant pro-
 277 cess $\rightarrow\rightarrow 4\ell$ was generated at the NLO accuracy level containing s-, t- and u-
 278 channel diagrams (Figure 1.4) using POWHEG V2 [28] and PYTHIA8, with the same
 279 configurations as for the Higgs signal. The dynamical renormalization and fac-
 280 torization scales have been chosed equal to m since the simulation covers a wide
 281 range of invariant masses of ZZ.

282 The $\rightarrow ZZ$ process (Figure 1.4) is simulated at leading order (LO) with MCFM
 283 [29, 30]. In order to have matching in spectra of $\rightarrow H \rightarrow ZZ$ as predicted at NLO
 284 by POWHEG, the showering of these samples has been performed with some dif-
 285 ferent PYTHIA8 settings. This only allows emissions only up to parton-level scale
 286 (“wimpy” shower).

287 The initial-state and final-state radiations are included for all events genera-
 288 tors, which provides hints for the existence of extra (hard) photons in the event.
 289 Interference between all amplitudes of process $H \rightarrow 4\ell$ are properly taken into
 290 consideration both in POWHEG and the JHUGEN. For the generators (at LO, NLO
 291 at NNLO) CTEQ6L [31], CT10 [32] and MSTW2008 [33] PDFs are used.

292 The generated events were interfaced to PYTHIA 6.4 Tune Z2* [34] which fol-
 293 lows CTEQ6L, to include the fact of the parton shower, underlying events and
 294 hadronization.

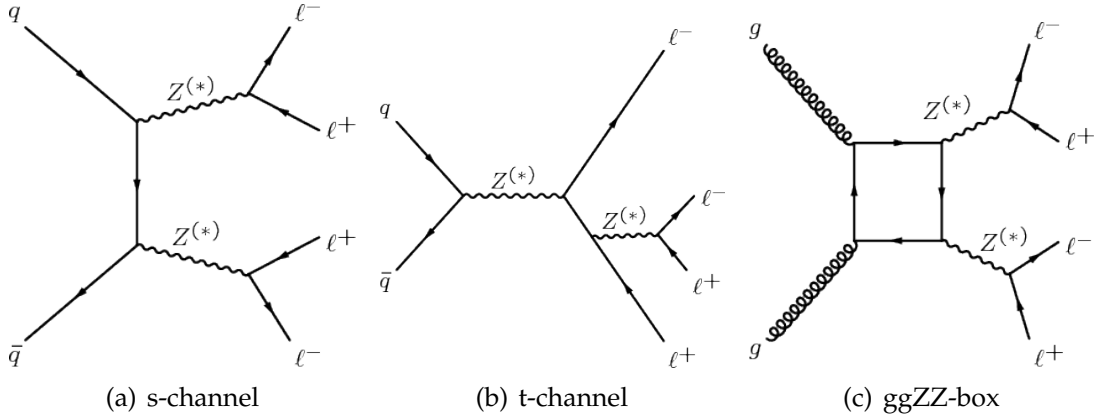


FIGURE 1.4: $t\bar{t} \rightarrow 4\ell$ process s- and t- channel fynman diagrams (a) and (b). $\rightarrow\rightarrow 4\ell$ process fynman diagram (c).

Processing of all generated events was made via a full simulation for CMS detector using [35] and their reconstruction was done by exploiting same algorithms used for the data analysis. Multiple proton-proton interactions are incorporated in the simulations by reweighting the pileup distribution to match the distribution of the observed data. While measuring the energy of jets and isolation of objects (using FASTJET technique) [36, 37, 38], the energy deposits from pileup interactions and underlying events are subtracted.

All the data-to-MC scale factors are computed by using a “T&P” technique [39], as described in Sections ?? and ??, and are used to correct simulated samples to compensate the differences of leptons’ selection efficiencies in the data and MC simulation. There is negligible dependence of the selection efficiencies used in the analysis over the primary vertices. More details are presented and discussed in Ref. [40].

Pileup Reweighting

For each year, corresponding simulation samples are reweighted to meet pileup distribution in the data. An example of the reweighting procedure for 2018 data is shown in Fig. 1.5.

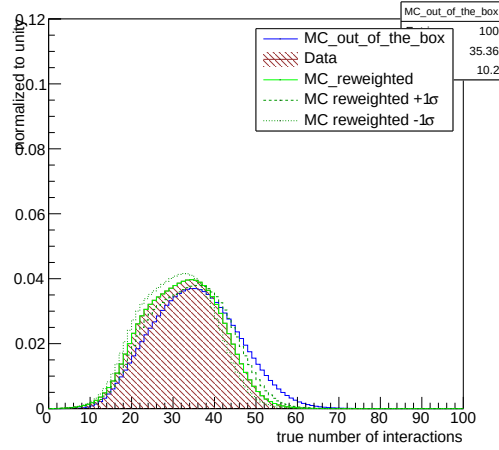


FIGURE 1.5: Pileup distribution in 2018 MC and Data shown before and after the application of PU weights. Up and down variations of 5% in the minimum bias cross section when calculating the weights are also shown.

1.2 Analysis Strategy

In order to minimize the effects of model dependent acceptances, measurements on Higgs cross sections (Inclusive and differential) are made in a fiducial region. Fiducial selection mimics the reconstruction level selection, which is optimized to detect a low mass Higgs (m_H 125) boson decaying to 4 leptons through a pair of Z bosons. Since in SM there are multiple production modes, the fiducial selection and measurement strategy are designed to be independent of how the Higgs boson is produced.

For this reason, the inclusion of isolation in the fiducial phase space definition is necessary to make the reconstruction efficiencies independent of fiducial phase space.

Furthermore, in sub-leading SM production modes there can be associated vector bosons which can decay to leptons, and therefore the reconstructed Higgs boson candidate may not be composed of the correct combination of the leptons. In order to retain the full information from an unbinned simultaneous fit of the signal and the background PDFs to the four-lepton distribution of invariant mass, it is necessary to treat such “wrong combination” events as a background.

Further signal-induced background can be formed due to imperfect correspondence between quantities at particle level and corresponding quantities at the

reconstruction level. Signal events which are from within fiducial phase space but fail the reconstruction are accounted in the fiducial efficiency (ϵ) and the expected contribution from events passing the reconstruction selections but are out of the fiducial phase space (f_{nonfid}) are referred to as “nonfiducial signal” events and are treated as a background, illustrated in Figure 1.6.

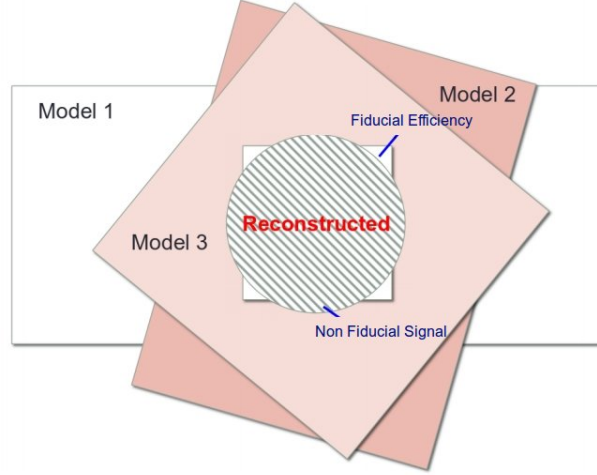


FIGURE 1.6: Fiducial efficiency(ϵ) and nonfiducial(f_{nonfid}) signal definition.

The effects of imperfect detector resolution can in general have a non-negligible effect on shape of the distribution of measured observables. For this reason, for differential cross section measurements of Higgs, a procedure to correct the detection efficiencies and resolution effects is applied. Throughout this document, this procedure will be referred to as the unfolding procedure, and the unfolded differential distributions will be referred to as distributions at the fiducial level. The reconstruction level selection, fiducial phase space definition, statistical procedure, and their model dependence will be described in the subsequent Sections.

1.2.1 Event Selection and Background Modeling

Event Selection

This dissertation is based on the analysis for which event selection for measurements are similar to what was used in pervious analyses on same topic and in the same final state [CMS:2018mmw, CMS:2017jkd, 40, 41]. The online event selection is based on having fired High Level Trigger paths as described in [sec:trigpaths].

Unlike in the Run I analysis, the trigger requirement does not depend on the selected final state: it is always the OR of all HLT paths. The reason is in Run II we will be targeting associated production modes that can come with additional leptons, thus improving trigger efficiency further. The events are required to have at least one primary vertex (PV) that fullfills the criteria of having large degree of freedom ($N_{PV} > 4$) with radius of PV being small ($r_{PV} < 2$ cm) and the collisions being restricted along z -axis ($z_{PV} < 24$ cm).

For the double lepton triggers, the requirement on minimum p_T for leading and the subleading lepton are imposed to be 17 and 8 GeV respectively; however, for triple electron triggers, it is 15, 8 and 5 GeV in the same order respectively.

1.2.2 ZZ Candidate Selection

The four-lepton candidates are built from what we call **selected leptons**, which are the tight leptons (defined in sections ?? and ??) that pass the $SIP_{3D} < 4$ vertex constraint and the isolation cuts (defined in sections ?? and ??), where FSR photons are subtracted as briefed in Section ?. A lepton cross cleaning is applied by discarding electrons which are within $\Delta R < 0.05$ of selected muons.

The construction and selection of four-lepton candidates proceeds according to the following sequence:

1. **Z candidates** are built from oppositely charged and same flavoured leptons (e^+e^- , $\mu^+\mu^-$) with additional requirement on dilepton mass as $12 < m_{\ell\ell(\gamma)} < 120 \text{ GeV}$. Z boson candidate is also required to include FSR photons (if any).
2. **ZZ candidates** are then built from non-overlapping Z boson candidates. The one who is reconstructed with $m_{\ell\ell}$ closer to the SM Z boson mass is regarded as Z_1 and other as Z_2 . Additional requierements on ZZ candidates are as under:
 - **Ghost removal** : Selected four leptons are required to have $\Delta R(\eta, \phi) > 0.02$ among each other.
 - **lepton p_T** : Among selected four leptons, two are required to fulfill $p_{T,i} > 20$ and $p_{T,j} > 10$.
 - **QCD suppression**: Dilepton inivariant mass is required to pass (regardless of flavor) $m_{\ell\ell} > 4$ for QCD induced J/Ψ veto.

- 381 • Z_1 **mass**: $40 \text{ GeV} < m_{Z1} < 120 \text{ GeV}$
- 382 • **four-lepton invariant mass**: $m_{4\ell} > 70$

383 3. Events containing one selected ZZ candidate at least should form the **signal**
 384 **region**.

385 1.2.3 Choice of best ZZ Candidate

386 Unlike in the Run I analysis, the best ZZ candidate is now chosen after all
 387 kinematic cuts, a change that allows to test other selection strategies for this can-
 388 didate choice. This is especially relevant for events with more than four selected
 389 leptons, having in mind search for Higgs boson production modes where associ-
 390 ated particles can decay to leptons, such as VH and ttH.

391 For fiducial cross section measurement, Z_1 is chosen with mass closest to the
 392 mass of SM Z boson mass while Z_2 is chosen from Z candidates whose lepton pair
 393 gives higher p_T sum.

394 Reconstruction of the jets is incumbent to explore the differential cross section
 395 measurements with respect to jets related observables. Jets are reconstructed with
 396 $p_T \geq 30 \text{ GeV}$ within $|\eta| \leq 4.7$, discussed in Section ???. To remove overlapping of
 397 jets overlap from leptons or selected photons, it is required to have at least $\Delta R >$
 398 0.5 where $\Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$ [42] with respect to any of the selected leptons
 399 and FSR photons candidates.

400 Background Modeling

401 The dominant and sub-dominant irreducible backgrounds for $H \rightarrow 4\ell$ process
 402 are the production of ZZ via annihilation of a $q\bar{q}$ pair and the gg fusion respectively.
 403 Other background is reducible which is of the instrumental nature and arises from
 404 the process where partial jets are misidentified as leptons. The main processes
 405 which contributes in this background are: production of jets in association with
 406 Z boson, production of the jets in association with WZ boson pair and the $t\bar{t}$ pair
 407 production. This background is as Z+X.

408 For fiducial cross section measurement for $H \rightarrow 4\ell$, both of irreducible back-
 409 grounds ($\rightarrow$ and $gg \rightarrow$) are estimated from Monte Carlo (MC) simulation as done
 410 earlier in Ref. [40]. In addition, following the study in Ref. [43], in these measure-
 411 ments the expected yield for the $gg \rightarrow$ background has been increased by assigning

the same ($m_{4\ell}$ dependent) k-factor as signal $H \rightarrow 4\ell$ process to leading order (LO) cross section of background. In the region ($105 < m_{4\ell} < 140$) this factor leads to an increase of expected $gg \rightarrow ZZ$ background by factor of ~ 2.66 . After inclusion of this k-factor, the $gg \rightarrow ZZ$ process is still a sub-dominant background in all measurements.

Reducible background (Z+X) is estimated using data. A tight-to-loose “fake rate” technique is used in the control regions of data by following Ref. [40]. Z+X background yields are determined using two methods:

- Opposite Sign (OS) method

TABLE 1.1: Number of observed candidates, expected background and the signal events passing all selections in mass range $m_{4\ell} > 70$ corresponding to three different years. Z+X is estimated by control regions in the data while ZZ and signal events are estimated using MC simulation.

Channel	4	4	22	4 ℓ
2016 (13 TeV)				
$\rightarrow$	$192.7^{+18.6}_{-20.1}$	$360.2^{+24.9}_{-27.3}$	$471.0^{+32.6}_{-35.7}$	$1023.9^{+68.9}_{-76.0}$
$gg \rightarrow$	$41.2^{+6.3}_{-6.1}$	$69.0^{+9.5}_{-9.0}$	$101.7^{+14.0}_{-13.3}$	$211.8^{+28.9}_{-27.5}$
+ X	$21.1^{+8.5}_{-10.4}$	$34.4^{+14.5}_{-13.2}$	$59.9^{+27.1}_{-25.0}$	$115.4^{+31.9}_{-30.1}$
Sum of backgrounds	$255.0^{+23.9}_{-25.1}$	$463.5^{+31.9}_{-33.7}$	$632.6^{+44.2}_{-46.1}$	$1351.1^{+85.8}_{-91.2}$
Signal ($m_H = 125$)	$12.1^{+1.3}_{-1.4}$	23.8 ± 2.1	30.3 ± 2.6	66.3 ± 5.6
Total expected	$267.1^{+24.9}_{-26.1}$	$487.4^{+33.1}_{-34.9}$	$662.9^{+45.7}_{-47.5}$	$1417.4^{+89.1}_{-94.3}$
Observed	293	505	681	1479
2017 (13 TeV)				
$\rightarrow$	$233.86^{+y.y}_{-z.z}$	$441.08^{+y.y}_{-z.z}$	$569.51^{+y.y}_{-z.z}$	$1244.45^{+y.y}_{-z.z}$
$gg \rightarrow$	$49.11^{+y.y}_{-z.z}$	$81.50^{+y.y}_{-z.z}$	$121.05^{+y.y}_{-z.z}$	$251.66^{+y.y}_{-z.z}$
+ X	$12.39^{+y.y}_{-z.z}$	$36.39^{+y.y}_{-z.z}$	$44.45^{+y.y}_{-z.z}$	$93.24^{+y.y}_{-z.z}$
Sum of backgrounds	$295.37^{+y.y}_{-z.z}$	$558.97^{+y.y}_{-z.z}$	$735.02^{+y.y}_{-z.z}$	$1589.35^{+y.y}_{-z.z}$
Signal ($m_H = 125$)	$13.85^{+y.y}_{-z.z}$	$28.64^{+y.y}_{-z.z}$	$35.55^{+y.y}_{-z.z}$	$78.04^{+y.y}_{-z.z}$
Total expected	$309.22^{+y.y}_{-z.z}$	$587.60^{+y.y}_{-z.z}$	$770.57^{+y.y}_{-z.z}$	$1667.39^{+y.y}_{-z.z}$
Observed	307	602	797	1706
2018 (13 TeV)				
$\rightarrow$	333^{+57}_{-53}	622^{+31}_{-44}	815 ± 73	1770^{+98}_{-101}
$gg \rightarrow$	$75.1^{+14.3}_{-13.5}$	$116.6^{+11.7}_{-12.8}$	176.9 ± 23.0	$368.5^{+29.5}_{-29.6}$
+ X	19.3 ± 7.2	50.8 ± 15.2	64.6 ± 15.6	134.7 ± 22.9
Sum of backgrounds	$428^{+59.2}_{-55.2}$	$790^{+36.4}_{-48.3}$	1057 ± 78.1	$2274^{+104.9}_{-107.7}$
Signal ($m_H = 125$)	$19.6^{+3.3}_{-3.1}$	$40.8^{+2.5}_{-2.9}$	50.7 ± 5.6	$111.1^{+6.9}_{-7.0}$
Total expected	$447^{+59.3}_{-55.2}$	$830^{+36.5}_{-48.4}$	1108 ± 78.3	$2385^{+105.1}_{-107.9}$
Observed	462	850	1130	2442

- Same Sign (SS) method

Both methods are found to be consistent with each other within the statistical uncertainties. First control region is attained with the requirement that two leptons do pass loose selections which does fulfill the criteria of 1 candidate (failing ID and isolation requirements), while the other two leptons in the event passes the criteria of 1 candidate. Thus, the sample is represented as "2 Prompt + 2 Fail" (2P2F) which is populated with the events which contain two prompt leptons (which are mostly Drell-Yan with small fraction of Z/γ^* and $t\bar{t}$ events) only. Second control region is attained by requiring that three leptons pass the final requirements. It is represented as "3 Prompt + 1 Fail" (3P1F) sample.

The reducible background's $m_{4\ell}$ shape is attained from the OS method by performing a fit of $m_{4\ell}$ distributions of 3P1F and 2P2F events independently by empirical analytical functional constructed by Landau [44] in Ref. [40]. Reducible background prediction in all three channels fitted together using this with total uncertainty indicated is shown in Figure ???. For differential measurements in $H \rightarrow 4\ell$, shape of reducible background is attained from templates built from the control regions in the data for each differential bin of the observable considered for measurement. This procedure is similar to what was used for spin-parity studies as described in the Ref. [40]. More details on this procedure of differential measurements of $H \rightarrow 4\ell$ will be given in Section 1.2.7.

The number of estimated signal and the background events, along with observed events in data passing all selections in mass range $m_{4\ell} > 70$ are listed in Table 1.1 for full Run 2 datasets.

Reconstructed four-leptons invariant mass distribution for whole Run 2 datasets and MC samples in full and high mass range is shownn in the Figure 1.7.

1.2.4 Definition of the Fiducial Volume

Fraction of all $H \rightarrow 4\ell$ events that get selected following the event selection procedure can change significantly for various production mechanisms of Higgs and for its distinct exotic models. In order to minimize the effects of model dependent acceptances, measurements on Higgs cross sections (Inclusive and differential) in $H \rightarrow 4\ell$ are made in a fiducial region. This fiducial phase space is defined so as to closely match with the experimental acceptance as defined from selections made at reconstruction level.

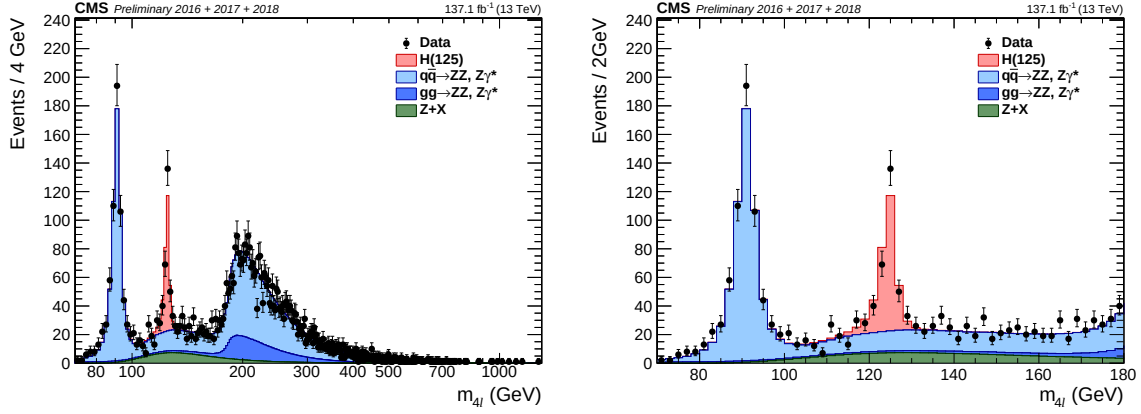


FIGURE 1.7: Invariant mass distributions of invariant mass of four-leptons using full Run 2 datasets in the full (left) and the low (right) mass range, along with predictions for Standard Model Higgs ($m_H = 125.0$ GeV), including $Z \rightarrow 4\ell$ resonance.

Requirement of fiducial phase space is minimum four leptons having $p_T > 7$ and $|\eta| < 2.5$ in case of electron, and $p_T > 5$ and $|\eta| < 2.4$ in case of muon. Besides, the leading and sub-leading leptons are required to have $p_T > 20$ and $p_T > 10$ respectively. The four-momenta of each lepton is calculated at the hard scattering level (PYTHIA status code = 3), before any final state radiation occurs. Each lepton isolation is calculated by adding p_T of all stable particles (PYTHIA status code = 1) within a cone size $\Delta R < 0.3$ around the lepton. All neutrinos and FSR photons are excluded from above summation. Then the lepton's relative isolation which is defined by ratio of this sum and p_T of considered lepton and is required to be less than 0.35, consistent with reconstruction level isolation requirement [40]. The requirement of isolation in definition of fiducial phase space is crucial to control the difference of signal selection efficiency among different models. The studies based on simulation samples show that signal efficiency can change up to 45% among different models without requiring lepton isolation at fiducial level as shown in Table ??, the maximum difference comes from $t\bar{t}H$ and $gg \rightarrow H$ production modes. This behavior is expected as $t\bar{t}H$ events has more hadronic activity around leptons as compared to $gg \rightarrow H$ events, so the cut at relative isolation kills more events in $t\bar{t}H$ than $gg \rightarrow H$.

Furthermore, fiducial phase space is defined by several other topological and kinematics selections. There must be two SFOS lepton pairs where Z_1 being the one with invariant mass closest to the SM Z boson and reconstructed with $40 \leq m(Z_1) \leq 120$ GeV. Z_2 is reconstructed by remaining SFOS pair of leptons within inv. mass range of $12 \leq m(Z_2) \leq 120$ GeV. In case of more than one Z_2 passing

the criteria, leptons pair with highest $\Sigma|p_T|$ is selected. To remove the overlapping between leptons, we ask for $\Delta R(\ell_i \ell_j) > 0.02$ for any $i \neq j$. To remove contamination from QCD low masses resonance, any opposite-charge pair of leptons is required to meet the condition $m(\ell^+ \ell'^-) \geq 4$. Finally, We require that all selected leptons must emerge from Higgs boson decay and invariant mass of four-leptons must fulfill $105 < m(4\ell) < 140$. Demand for $m(4\ell)$ helps to control the contamination from off-shell gg fusion production which is of considerable size, capable to increase the total cross section by a few percent [45, 46]. The requirements that defines fiducial phase space are also gathered in Table 1.2.

TABLE 1.2: Overview of all requirements utilized to define the fiducial phase space.

Requirements on $H \rightarrow 4\ell$ fiducial phase space	
Lepton kinematics and the isolation	
p_T of the leading lepton	$p_T > 20 \text{ GeV}$
p_T of sub-leading lepton	$p_T > 10 \text{ GeV}$
Additional electrons (muons) p_T	$p_T > 7 \text{ (5) GeV}$
Pseudorapidity of electrons (muons)	$ \eta < 2.5 \text{ (2.4)}$
Sum of the scalar p_T of all the stable objects within $\Delta R < 0.3$ from the lepton	$< 0.35 p_T$
Event topology	
From leptons satisfying criteria above, minimum two SFOS lepton pairs	
Invariant mass of Z_1 candidate	$40 < m(Z_1) < 120$
Invariant mass of Z_2 candidate	$12 < m(Z_2) < 120$
Separation between selected four leptons	$\Delta R(\ell_i \ell_j) > 0.02$
Invariant mass of any OS lepton pair	$m(\ell_i^+ \ell_j^-) > 4$
Invariant mass of selected four leptons	$105 < m_{4\ell} < 140$

It has been established based on simulation samples that with including isolation for the events passing fiducial phase space definition reconstruction efficiencies faintly depends on Higgs properties and its production modes. The remaining model dependence will be quoted as a separate uncertainty as explained in Section 1.3. Part of the signal events passing fiducial phase space definition ($\mathcal{A}_{\text{fid}}$), and reconstruction efficiencies (ϵ) for SM production modes are registered in Table 1.3.

TABLE 1.3: Summary of different SM signal models. The MC samples are from full Run 2 yearly productions (2016, 2017 and 2018) assuming $m_H = 125.0$. Luminosity averaged fiducial acceptance ($\mathcal{A}_{\text{fid}}$) defined the fraction of events passing fiducial phase space selections, reconstruction efficiency (ϵ) represented by the ratio of reconstructed to accepted events. Also, f_{nonfid} which are reconstructed events but does not pass fiducial phase space definition at generator level. Values are obtained using 13 TeV signal models and the uncertainties shown are only statistical uncertainties.

Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual production modes of Higgs boson				
gg $\rightarrow$ H (POWHEG)	0.402 ± 0.001	0.592 ± 0.002	0.053 ± 0.001	0.624 ± 0.002
VBF (POWHEG)	0.444 ± 0.002	0.605 ± 0.003	0.043 ± 0.001	0.631 ± 0.003
WH (POWHEG+MINLO)	0.325 ± 0.002	0.588 ± 0.003	0.075 ± 0.002	0.632 ± 0.004
ZH (POWHEG+MINLO)	0.340 ± 0.003	0.594 ± 0.005	0.081 ± 0.004	0.643 ± 0.006
ttH (POWHEG)	0.314 ± 0.003	0.585 ± 0.006	0.169 ± 0.006	0.684 ± 0.007

1.2.5 Signal Model and Statistical Procedure

The purpose is to extract integrated and differential fiducial cross sections after corrections of detector effects, resolution and systematic biases. To achieve this target, the corrections are extracted by using simulated samples and then encoded in the parametrization of $m_{4\ell}$ spectra at the reconstruction level. Then we directly extract the parameter of interest (σ_{fid}) by performing the maximum likelihood fit of background and signal parametrization to the observed $m_{4\ell}$ distribution. All the systematic uncertainties in the measurements are encoded as nuisance parameters, which are profited properly in fit procedure. In order to obtain results, we used asymptotic approach [47] with test statistics that is based on profile likelihood ratio [48]. Maximization of likelihood is performed by fitting simultaneously, in each bin, the signal cross section at the fiducial level with all reconstruction bins and all three final states. The procedure applied for $H \rightarrow 4\ell$ fiducial (integrated and differential) cross section measurements will be given in Sections 1.2.6 and 1.2.7. Finally, the unfolding procedure to extract fiducial cross section kinematic observables' bins at the fiducial level, corrected for the detection efficiency and resolution effects, is performed as discussed in Ref. [49] and similar to what is adopted for $H \rightarrow \gamma\gamma$ decay channel [50].

Double Crystal Ball function is used for shape description of signal resonant contribution ($\mathcal{P}_{\text{res}}(m_{4\ell})$) [40] as is given below, with the yield proportional to the

parameter of interest (σ_{fid}).

$$f_{dCB}(x; \alpha, n, s, \sigma) = N \left\{ \begin{array}{ll} e^{-\frac{(x-s)^2}{2\sigma^2}} & \text{if } -\alpha_L < \frac{x-s}{\sigma} < \alpha_R \\ A_L(B_L - \frac{x-s}{\sigma}^{-n_L}) & \text{if } \frac{x-s}{\sigma} \leq -\alpha_L \\ A_R(B_R + \frac{x-s}{\sigma}^{-n_R}) & \text{if } \frac{x-s}{\sigma} \geq \alpha_R \end{array} \right\} \quad (1.1)$$

$$A_L = \left(\frac{n_L}{\alpha}\right)^{n_L} \times e^{-\frac{|\alpha_L|^2}{2}}$$

$$A_R = \left(\frac{n_R}{\alpha}\right)^{n_R} \times e^{-\frac{|\alpha_R|^2}{2}}$$

$$B_L = \frac{n}{\alpha_L} - |\alpha_L|$$

$$B_R = \frac{n}{\alpha_R} - |\alpha_R|$$

N is the normalization constant.

In case of $(Z)H$, and $t\bar{t}H$ modes of Higgs production, the event selection procedure will occasionally select 4ℓ candidates where minimum one lepton is produced from associated W or Z bosons or jets, and not from $H \rightarrow 4\ell$ process. Fraction of such signal events in $105 < m(4\ell) < 140$ mass range mounts to about 4%, 18% and 22% for WH, ZH and $t\bar{t}H$ modes of production, respectively. These events have a broad invariant $m_{4\ell}$ mass distribution that depends on the exact production mechanism. The shape of the contribution by this background, $\mathcal{P}_{\text{comb}}(m_{4\ell})$, is described a Landau distribution with parameters extracted from simulations.

f_{nonfid} represents the ratio of the events reconstructed from out of fiducial region to those reconstructed from within fiducial volume. For a particular bin j at reconstruction level, it is given as:

$$f_{\text{nonfid}}^j = \frac{N(\text{j}_{\text{reco}} \&\& \text{not fid.})}{N(\text{j}_{\text{reco}} \&\& \text{fid.})}. \quad (1.2)$$

Such events arise from imperfect detector resolution which causes the differences for quantities calculated at generated and reconstructed levels. Fraction of these events are also treated as background and will be referred "non-fiducial signal" throughout this disseration. It has been demonstrated with MC samples that these events can be described using same shape as resonant component. The values of such fraction are extracted from simulation repeatedly for SM. This fraction varies among different production modes of Higgs e.g. it is 4% for the $gg \rightarrow H$ and 3%

for ttH production mode. The values for each model are listed in the Tables ??, ?? and ?? for 2016, 2017 and 2018 respectively.

To make comparison with the theoretical predictions, measurements need corrections for resolution of detector and effects of efficiency. To unfold reconstruction-level distribution to the distribution before all detector effects, the efficiency matrix (ϵ) is constructed using the signal MC samples. The efficiency matrix (ϵ) provides the probability of an event that was generated in i th bin at the fiducial level and is reconstructed in j -th bin as:

$$\epsilon_{i,j} = \frac{N(j_{\text{reco}} \&\& i_{\text{gen}})}{N(i_{\text{gen}})}, \quad (1.3)$$

where $N(i_{\text{gen}})$ is total number of the events in i th bin at fiducial level and $N(j_{\text{reco}} \&\& i_{\text{gen}})$ is total number events which i th bin at the fiducial level but reconstructed in j th bin. For integrated measurements, where we have only one bin, fiducial efficiency is about 65% which can change upto 7% for other signal models. For differential measurements, it is described by detector response matrix for each observable. Figure ?? shows the efficiency matrix for $p_T(H)$ measurement using different SM production MC samples. The efficiency matrix is dominated by diagonal elements, but in general it can have non-negligible off-diagonal elements specially for jet related observables due to jet energy scale uncertainty. In case of differential variable, jet multiplicity, neighbour elements of diagonal varies from 3% to 21% as shown in Figures ??-?? in the Appendix ??.

The estimated number events for a studied differential observable in a final state f and in the defined bin i are described in terms of $m_{4\ell}$ given by:

$$\begin{aligned} N_{\text{obs}}^{f,i}(m_{4\ell}) &= N_{\text{fid}}^{f,i}(m_{4\ell}) + N_{\text{nonfid}}^{f,i}(m_{4\ell}) + N_{\text{comb}}^{f,i}(m_{4\ell}) + N_{\text{bkg}}^{f,i}(m_{4\ell}) \\ &= \sum_j \epsilon_{i,j}^f \cdot \left(1 + f_{\text{nonfid}}^{f,i}\right) \cdot \sigma_{\text{fid}}^{f,j} \cdot \mathcal{L} \cdot \mathcal{P}_{\text{res}}(m_{4\ell}) \\ &\quad + N_{\text{comb}}^{f,i} \cdot \mathcal{P}_{\text{comb}}(m_{4\ell}) + N_{\text{bkg}}^{f,i} \cdot \mathcal{P}_{\text{bkg}}(m_{4\ell}). \end{aligned} \quad (1.4)$$

Where $N_{\text{fid}}^{\text{f},i}(m_{4\ell})$, $N_{\text{nonfid}}^{\text{f},i}(m_{4\ell})$, $N_{\text{comb}}^{\text{f},i}(m_{4\ell})$ and $N_{\text{bkg}}^{\text{f},i}(m_{4\ell})$ denote the resonant fiducial signal, the nonfiducial resonant, the combinatorial contribution of fiducial signal, and background contributions in a defined bin i as functions of $m_{4\ell}$ (four lepton invariant mass), respectively. Then, probability density functions corresponding to the resonant (fiducial and the nonfiducial) signal, the combinatorial contributions of fiducial signal, and contribution from the background are $\mathcal{P}_{\text{res}}(m_{4\ell})$, $\mathcal{P}_{\text{comb}}(m_{4\ell})$ and $\mathcal{P}_{\text{bkg}}(m_{4\ell})$ respectively. $\epsilon_{i,j}^{\text{f}}$ is the response matrix that connects reconstructed level events in a defined bin i with fiducial events in the bin j for studied differential observable. The f_{nonfid}^i fraction represents the ratio of nonfiducial signal to the fiducial signal contributions in a defined bin i at reconstruction level. $\sigma_{\text{fid}}^{\text{f},j}$ is the parameter of interest that represents signal cross section for a final state f in a defined bin j of fiducial region.

1.2.6 Extraction of Integrated $\text{H} \rightarrow 4\ell$ Fiducial Cross Section

Signal Model and Statistical Procedure

For the integrated or inclusive cross section measurement within the defined region fiducial phase space, we follow a general procedure of differential cross sections measurement as outlined in Section 1.2.5, treating it as a special case where the number of considered bins is one.

Table 1.4 shows the reconstruction efficiency (ϵ) for fiducial events, while Table 1.5 shows the ratio (f_{nonfid}) of events reconstructed from out of fiducial region and the events reconstructed within fiducial volume, per final state for different Standard Model signal models. The primary source of the “nonfiducial signal” contributions (described with the factor f_{nonfid}) is the imperfect correspondence of isolation at particle and reconstruction levels, as discussed at the beginning of Section 1.2. In a similar way, fraction of the signal events in mass range 105.0–140.0 where minimum one lepton selected does not originate from the Higgs boson decay are shown in Table 1.6.

1.2.7 Extraction of Differential $\text{H} \rightarrow 4\ell$ Fiducial Cross Sections

Kinematic Observables

Measurements on differential cross section which are performed as function of the kinematic properties of four-lepton system and as well as jet multiplicity are

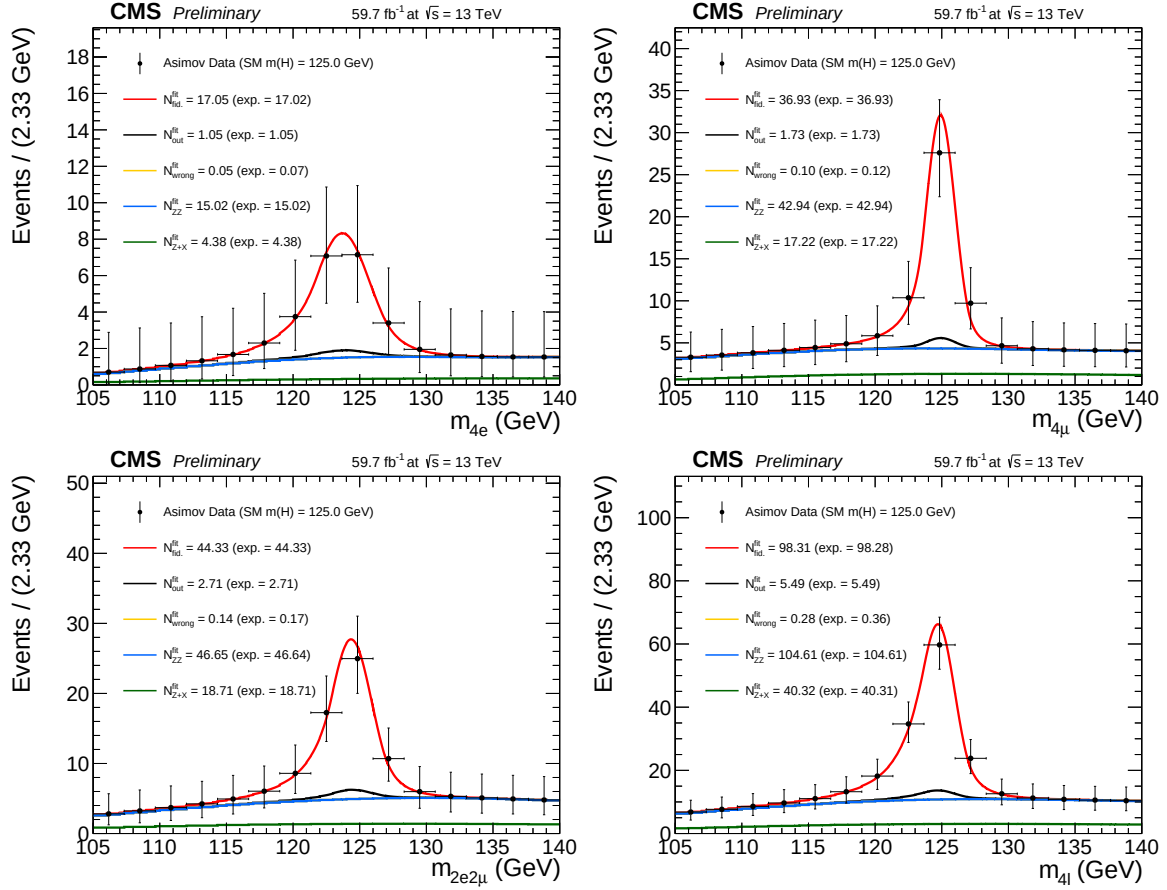


FIGURE 1.8: Inclusive four lepton mass distributions for an Asimov Dataset generated using SM cross section values and SM efficiencies where the $gg \rightarrow H$ prediction is from POWHEG+JHUGEN and resulting fitted values of the signal and background pdfs in different final states at $\sqrt{s} = 13$ TeV with 2018 data.

TABLE 1.4: Reconstruction efficiency (ϵ) for fiducial events per final state for different Standard Model signal models at $\sqrt{s} = 13$ TeV with 2018 data.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg $\rightarrow$ H (POWHEG)	0.404 ± 0.002	0.799 ± 0.002	0.572 ± 0.002	0.593 ± 0.001
VBF (POWHEG)	0.431 ± 0.003	0.805 ± 0.002	0.586 ± 0.002	0.608 ± 0.002
WH (POWHEG+MINLO)	0.415 ± 0.004	0.780 ± 0.003	0.580 ± 0.003	0.593 ± 0.002
ZH (POWHEG+MINLO)	0.428 ± 0.006	0.780 ± 0.005	0.581 ± 0.005	0.597 ± 0.003
ttH (POWHEG)	0.429 ± 0.006	0.745 ± 0.005	0.573 ± 0.005	0.585 ± 0.003

TABLE 1.5: Ratio of events reconstructed from out of fiducial volume and the events reconstructed within the fiducial region (f_{nonfid}) per final state in different Standard Model signal models at $\sqrt{s} = 13$ TeV with 2018 data.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg $\rightarrow$ H (POWHEG)	0.060 ± 0.002	0.046 ± 0.001	0.060 ± 0.001	0.055 ± 0.001
VBF (POWHEG)	0.050 ± 0.002	0.037 ± 0.001	0.046 ± 0.001	0.043 ± 0.001
WH (POWHEG+MINLO)	0.098 ± 0.004	0.058 ± 0.002	0.080 ± 0.002	0.075 ± 0.001
ZH (POWHEG+MINLO)	0.105 ± 0.007	0.071 ± 0.004	0.091 ± 0.004	0.087 ± 0.002
ttH (POWHEG)	0.278 ± 0.012	0.127 ± 0.005	0.193 ± 0.006	0.185 ± 0.004

TABLE 1.6: Signal event fraction in the mass range 105.0–140.0 where minimum one lepton selected is originating from Higgs decay at $\sqrt{s} = 13$ TeV with 2018 data.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg $\rightarrow$ H (POWHEG)	0.002	0.002	0.002	0.002
VBF (POWHEG)	0.002	0.003	0.003	0.003
WH (POWHEG+MINLO)	0.036	0.031	0.042	0.037
ZH (POWHEG+MINLO)	0.199	0.207	0.212	0.208
ttH (POWHEG)	0.175	0.156	0.164	0.163

unfolded to the particle level. In this dissertation, mainly the kinematic properties of four-lepton system are exploited and the results are reported for transverse momentum and rapidity of Higgs or the four lepton system($p_T(\text{H})$ or $p_T(4\ell)$, $|y(\text{H})|$ or $|y(4\ell)|$), $N(\text{jets})$, and $p_T(\text{jet})$ which probe the production mechanism. For each kinematic or differential observable, selection procedure of the binning boundaries has been optimized in order to have a similar expected measured cross-section for every bin.

Background Model

In order to perform the differential measurements, the background is to be subtracted at the reconstruction level for each bin of the considered observable. In general, the background mass spectrum shape is not the same in all bins of the considered observable because the background does not have a resonance structure. The mass spectrum also can have non-negligible correlation with the observables.

In case of the irreducible backgrounds, the four-lepton mass spectra are modeled as the $m_{4\ell}$ template shapes using simulated events for each bin of the observable considered for the measurement. Similarly, in case of the reducible backgrounds the templates are built from the control regions in the data. This procedure is similar to the one described and used in the Ref. [40].

In each differential bin of $p_T(4\ell)$ fractions of background events are presented in the Table 1.7.

TABLE 1.7: Estimated fraction of background events in each bin of $p_T(4\ell)$, for each final state using 2018 samples.

Background	final state	0 – 15	15 – 30	30 – 45	45 – 80	80 – 120	120 – 200	200 – 13000
qqZZ	$2e2\mu$	0.483668	0.239196	0.130151	0.105276	0.0266332	0.011809	0.00326633
qqZZ	$4e$	0.459665	0.234399	0.136986	0.114916	0.0388128	0.00989346	0.00532725
qqZZ	4μ	0.492632	0.241203	0.110977	0.109173	0.0297744	0.0129323	0.00330827
ggZZ	$2e2\mu$	0.310446	0.309859	0.191901	0.168134	0.0187793	0.000880282	0.0
ggZZ	$4e$	0.3085	0.306332	0.19247	0.17433	0.0169994	0.00136908	0.0
ggZZ	4μ	0.321332	0.319311	0.194863	0.150987	0.0128703	0.000638196	0.0
Z+X (CR)	$4l$	0.0992521	0.211153	0.209307	0.305327	0.120303	0.0483797	0.00627828

Signal Model and Statistical Procedure

The fiducial differential cross section is extracted in the bins of kinematic observables at the fiducial level following the procedure outlined in Section 1.2.5. For signal yields, mass spectrum of signal is built in each cell of the response matrix using a Double Crystal Ball (DCB) function, similar to what was done in Ref. [40]. In practice, the same parameters of the CB function are used in each cell and the systematic uncertainties assigned on the energy scale and resolution generally cover discrepancy of signal mass spectrum among each cell.

Examples of the fitted signal line shapes for differential bins of $p_T(\text{H})$ from $gg \rightarrow \text{H}$ (POWHEG+JHUGEN 125 2018 sample) in the individual final states are shown in Figure 1.9, and examples of the efficiency matrices for $p_T(\text{H})$ for the $2e2\mu$ final state are shown in Fig 1.10. Many more examples can be seen in Appendix ??.

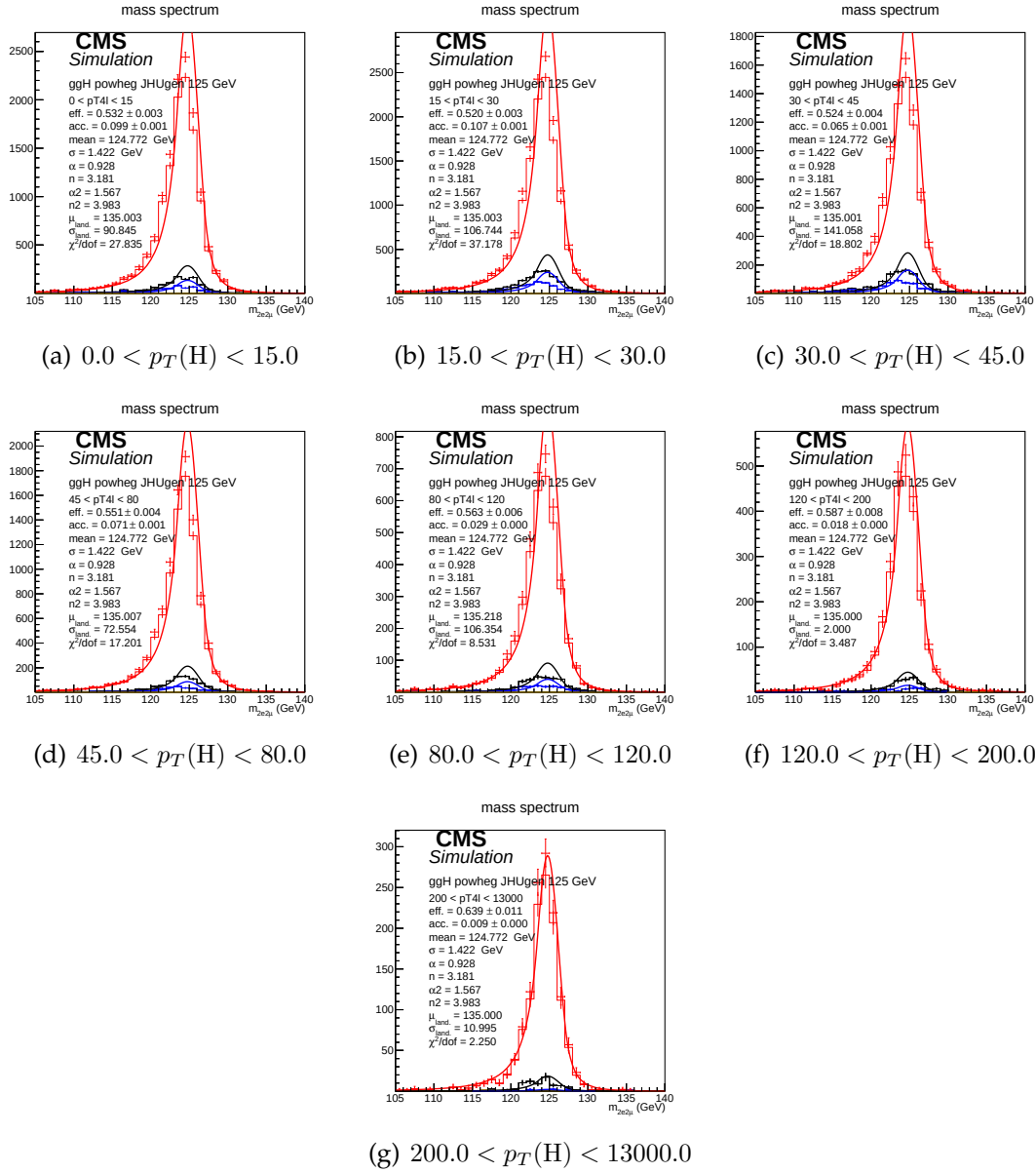


FIGURE 1.9: Distributions of $m(4\ell)$ in $2e2\mu$ final state for the $gg \rightarrow H$ production mode from POWHEG+JHUGEN in different bins of $p_T(H)$.

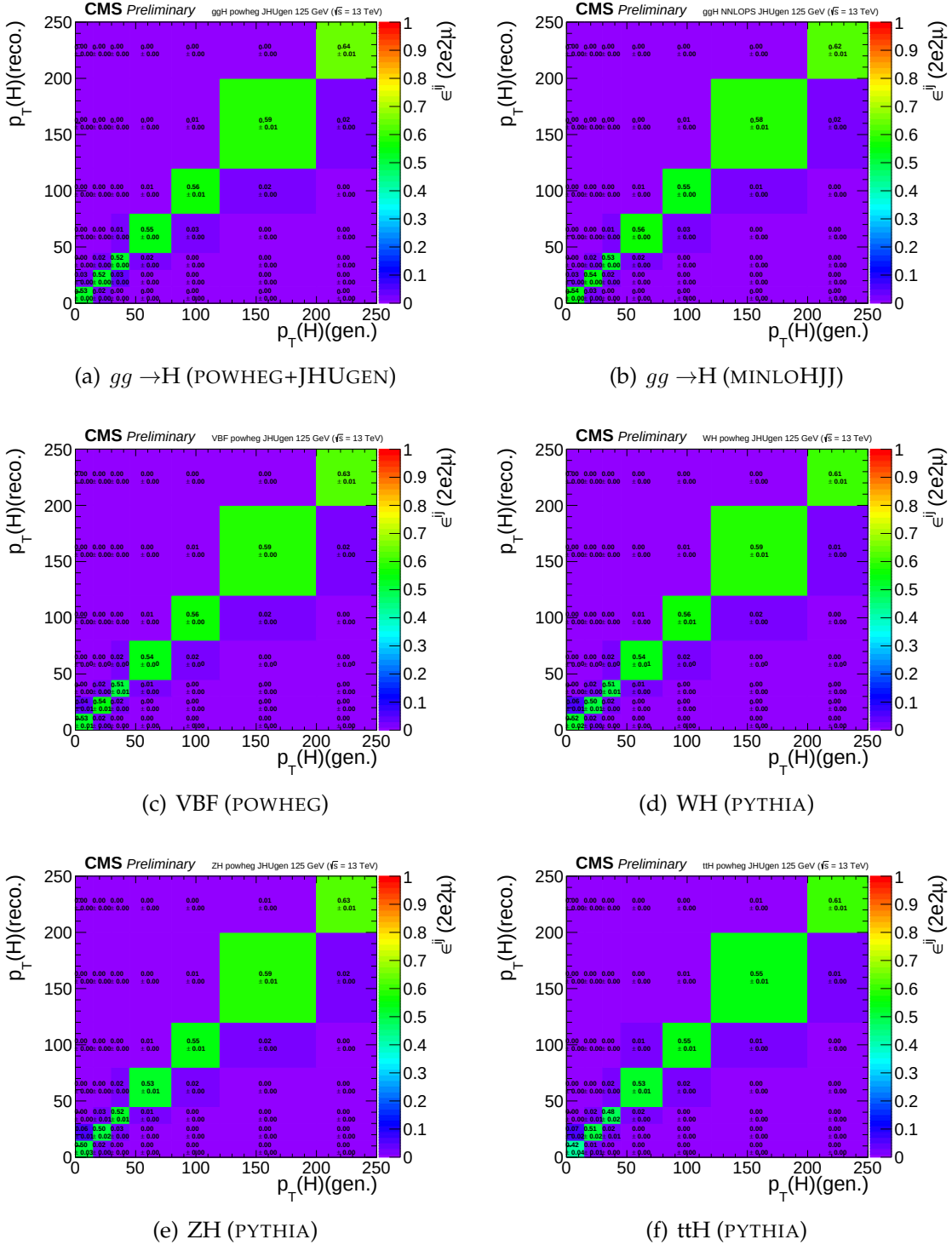


FIGURE 1.10: Efficiency matrices for $p_T(H)$ for different SM production modes in the $2e2\mu$ final state using 2018 samples.

621 Figure 1.11 shows the four lepton mass distributions for an Asimov dataset
622 generated using SM cross section values and efficiencies from the $gg \rightarrow H$ produc-
623 tion mode from POWHEG+JHUGEN, and resulting fitted values of PDFs of signal
624 and background for different bins of $p_T(H)$ (all final states combined).

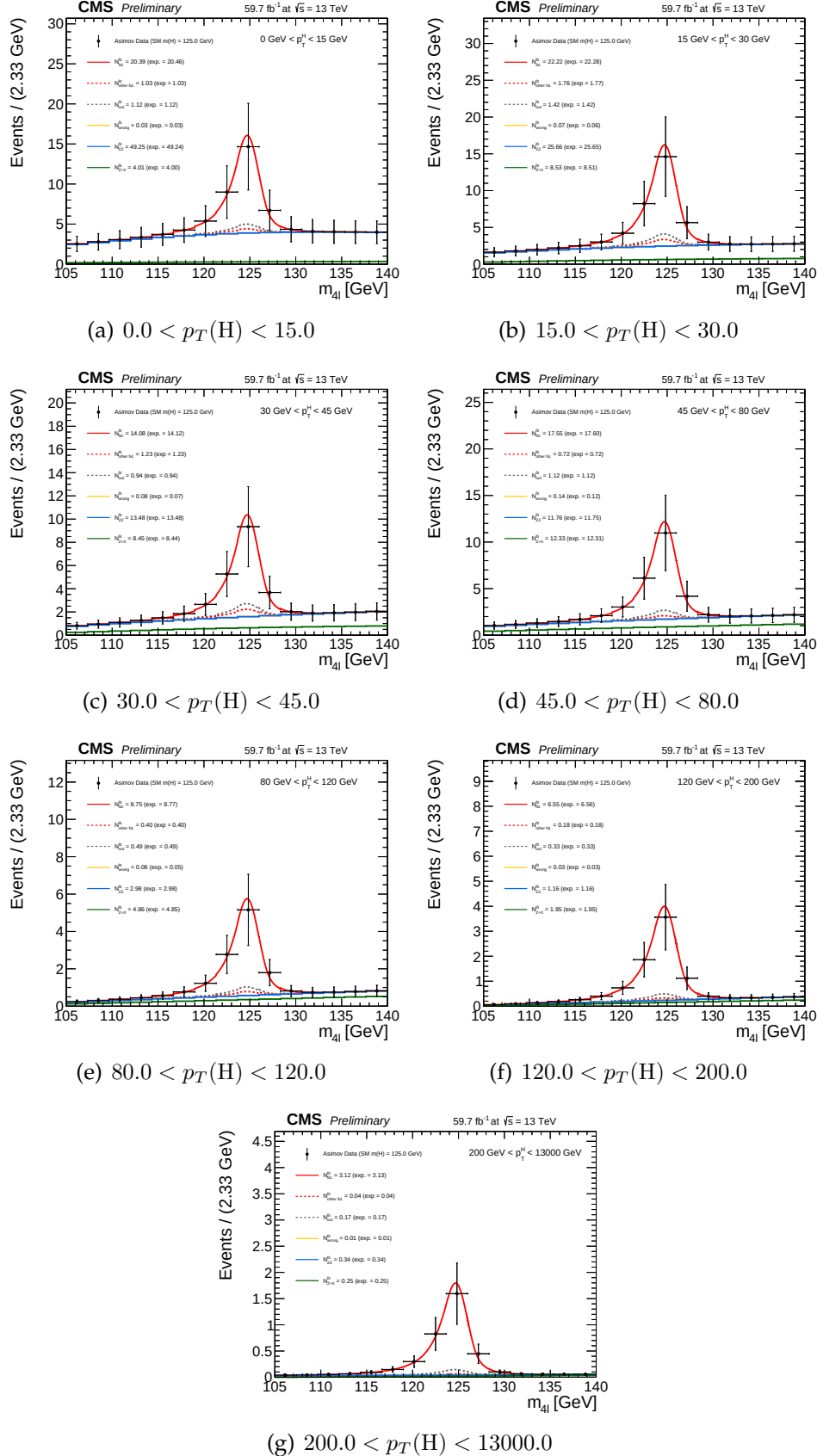


FIGURE 1.11: Four lepton mass distributions for an Asimov dataset generated using SM cross section values and SM efficiencies where the $gg \rightarrow H$ prediction is from POWHEG+JHUGEN and resulting fitted values of PDFs of signal and background for different bins of $p_T(H)$ (all final states combined).

1.3 Systematic Uncertainties

1.3.1 Experimental uncertainties sources

Main experimental uncertainties affecting signal and background are uncertainty on integrated luminosity (from 2.3% to 2.5%, depending on the year of data taking) and the uncertainty on efficiencies of lepton reconstruction and identification (ranging from 2.5–16.1% in 4μ and $4e$ channels, respectively). Uncertainties on estimation of reducible background, described in Section 1.2.3, originating from the background composition and misidentification rate uncertainty vary for different final states. Mismatch in composition of backgrounds between different samples where the fake rate is derived and where it is applied accounts between 24 and 43%.

Lepton energy scale uncertainty is assessed by propagating down to the four-leptons invariant mass the uncertainty associated the lepton momentum corrections on each individual lepton. It is estimated by exploiting $Z \rightarrow \ell\ell$ invariant mass distributions both in data and the simulation. Events are categorized on the basis p_T and η of either (randomly determined) of two leptons and integrating on the other lepton. The dilepton mass distributions are then fit by parameterization of a Breit-Wigner convoluted with a DCB function. The uncertainty is treated as uncorrelated (all up or all down). The new four-leptons invariant mass obtained this way is then fitted with only the mean value floating. Difference between new mean value and default one gives an estimation of impact of lepton energy scale. Values of the differences are reported in the Table 1.8, together with estimation of the impact of lepton momentum resolution, following a similar procedure.

TABLE 1.8: Difference (in %) between the nominal mean and the width of four-leptons or Higgs invariant mass distribution and those where the lepton momentum scale and resolution uncertainties have been propagated.

	$4e$	4μ	$2e2\mu$
Scale Difference (%)	0.087	0.049	0.069
Resolution Difference (%)	17	12	17

The results obtained this way suggest that the uncertainties from 2016 (20% on the resolution, 0.3% for the 4 channel for scale) are still valid.

1.3.2 Theoretical uncertainties

Estimation of the theoretical uncertainties affecting signal and the background include uncertainties from factorization and renormalization scale is done by varying the two scales 0.5-2 times the nominal values with their ratio kept in between 0.5-2. In the case of ggF production mode, the QCD scale uncertainty is broke down to 9 different sources. Other theoretical uncertainty source is PDF set choice which is determined RMS of the variation when different replias of default NNPDF set is used. Additional uncertainty on the background which is 10% on K factor applied on $ggZZ$ prediction is considered as described in Section 1.2.3. Systematic uncertainty of about 2% on the branching ratio of $H \rightarrow 4\ell$ is only applicable to signal yield. Additional sources are the imprecise jet energy scale and resolution knowledge (from 2% to 22%), the efficiency on b-tagging and the light quarks (*i.e.* u, d, s, c) and gluon jet mistag rate. While cross section measurement, signal cross section uncertainties arising from theoretical sources are removed.

TABLE 1.9: A chart of experimental systematic uncertainties in the $H \rightarrow 4\ell$ cross section measurements of 2016 data.

Relative systematic uncertainties	
Uncertainties from common experimental sources	
Luminosity	2.6 %
Lepton selection efficiencies	2.5 – 9 %
Uncertainties related to background	
Reducible background (Z+X)	36 – 43 %
Uncertainties related to Signal	
Lepton energy scale	0.04 – 0.3 %
Lepton energy resolution	20 %

1.3.3 Systematic uncertainties treatment when combining yearly measurements on data

In combination of three data taking periods, theoretical uncertainties as well as the experimental ones related to leptons or jets are treated as correlated while all other ones from experimental sources are taken as uncorrelated.

TABLE 1.10: A chart of experimental systematic uncertainties in the $H \rightarrow 4\ell$ cross section measurements of 2017 data

Relative systematic uncertainties	
Uncertainties from common experimental sources	
Luminosity	2.3 %
Lepton selection efficiencies	3. – 12.5 %
Uncertainties related to background	
Reducible background fake rate variation (Z+X)	6 – 32 %
Uncertainties related to Signal	
Lepton energy scale	0.04 – 0.3 %
Lepton energy resolution	20 %

TABLE 1.11: TO BE UPDATED. COPY PASTE NUMBERS OF 2017 FOR NOW. A chart of experimental systematic uncertainties in the $H \rightarrow 4\ell$ cross section measurements of 2018 data

Relative systematic uncertainties	
Uncertainties from common experimental sources	
Luminosity	2.3 %
Lepton selection efficiencies	3. – 12.5 %
Uncertainties related to background	
Reducible background fake rate variation (Z+X)	6 – 32 %
Uncertainties related to Signal	
Lepton energy scale	0.04 – 0.3 %
Lepton energy resolution	20 %

TABLE 1.12: Table theory uncertainties in the $H \rightarrow 4\ell$ Inclusive cross section measurements

Summary of inclusive theory uncertainties	
$\text{BR}(H \rightarrow \rightarrow 4\ell)$	2 %
QCD scale ($\rightarrow$)	+3.2/-4.2 % %
PDF set ($\rightarrow$)	+3.1/-3.4 %
Electroweak corrections ($\rightarrow$)	± 0.1 %

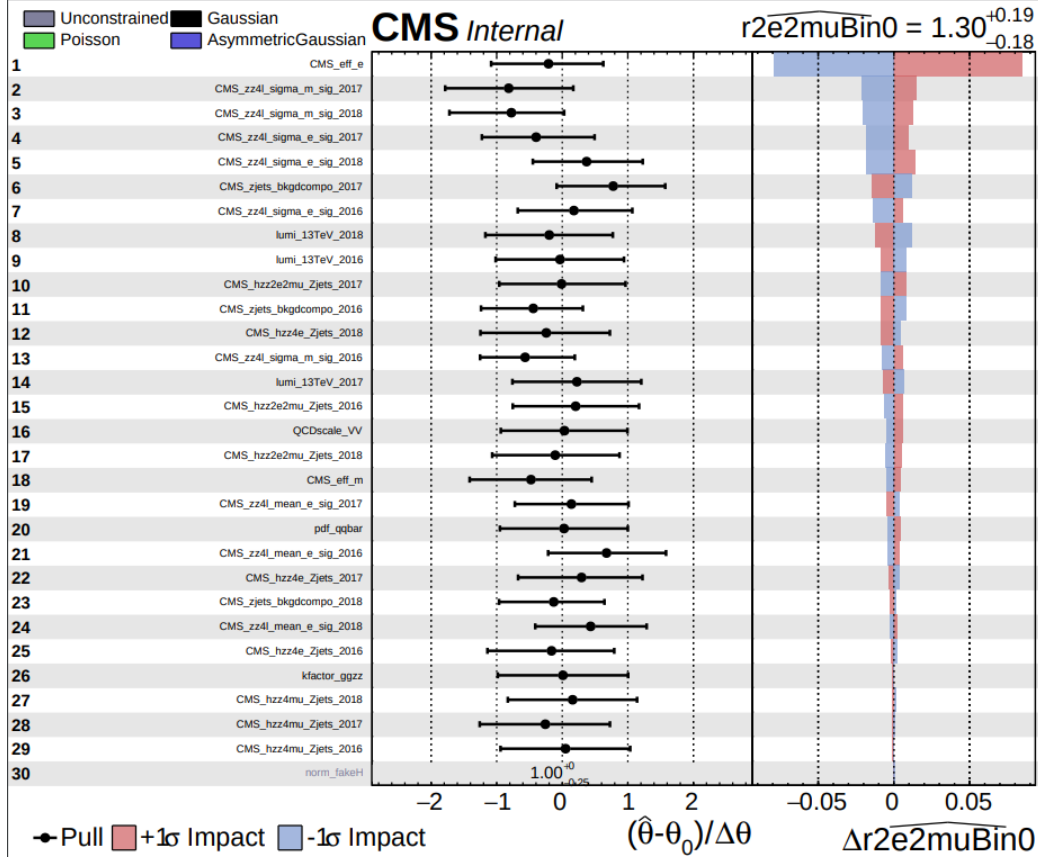


FIGURE 1.12: Pulls. Overall impact of different uncertainties sources on the inclusive measurement ($2e2\mu$ final state)

1.4 Results

1.4.1 Measurement of Integrated Fiducial Cross Section

Resulting fits to the inclusive $m(4\ell)$ distributions (extracted in $m_{4\ell}$ range *viz* 105 GeV to 140 GeV) for measurement of integrated $H \rightarrow 4\ell$ fiducial cross section in different final states, are shown in Figure 1.13. Extracted inclusive cross sections as function of colliding center of mass energy has been shown in the Figure 1.14(a) while the extracted integrated cross section results for different final states at 13 TeV corresponding to different years of data taking are shown in Figure 1.14(b). Systematic on the measurement is represented by red bar, whereas black error bar represents total uncertainties, summed from the individuals in quadrature. Contributions of the sub-dominant component of the signal i.e. $XH = VBF + VH + t\bar{t}H$ is indicated in green separately. These results are also given in Tables 1.13 and

1.14.

The inclusive $H \rightarrow 4\ell$ measurements are compared with SM NNLO theoretical estimations where acceptance of dominant $gg \rightarrow H$ contribution is estimated using POWHEG at $\sqrt{s}=13$ and HRES [51, 52] at $\sqrt{s}=7$ and 8. Total gluon fusion cross section and associated uncertainty has been taken from Ref. [53]. Definition of fiducial volume for $\sqrt{s}=6-9$ uses the lepton isolation definition from Ref. [54], while for $\sqrt{s}=12-14$ the definition described in the text is used.

When extracting the integrated $H \rightarrow 4\ell$ fiducial cross section, the parameter of interest is total fiducial cross section and the fractions of $4e$, 4μ , and $2e2\mu$ is floated in the fit. When extracting the fiducial cross section in the final states, three parameters of interest are the cross sections of three individual final states.

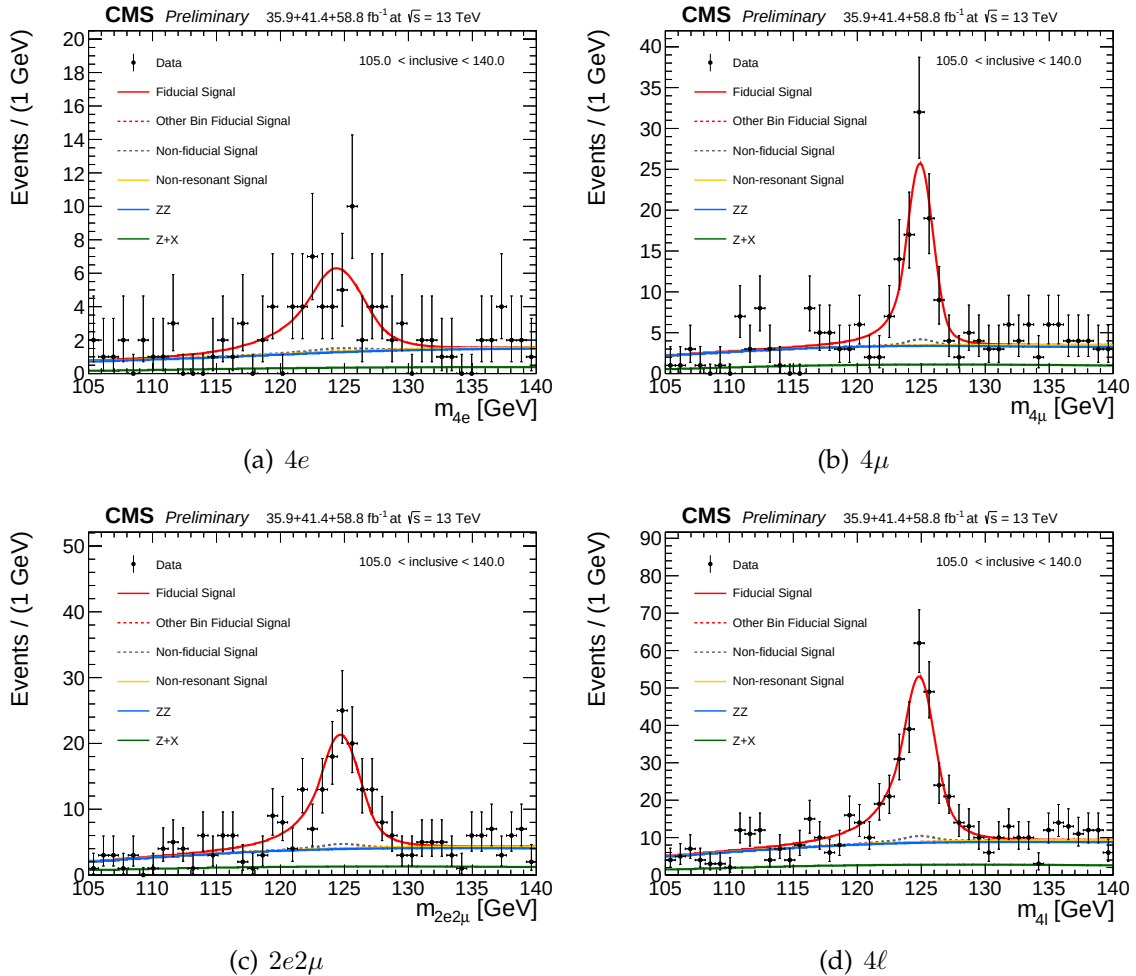


FIGURE 1.13: Inclusive four lepton invariant mass distributions in data and resulting fitted values of signal and the background PDFs in different final states.

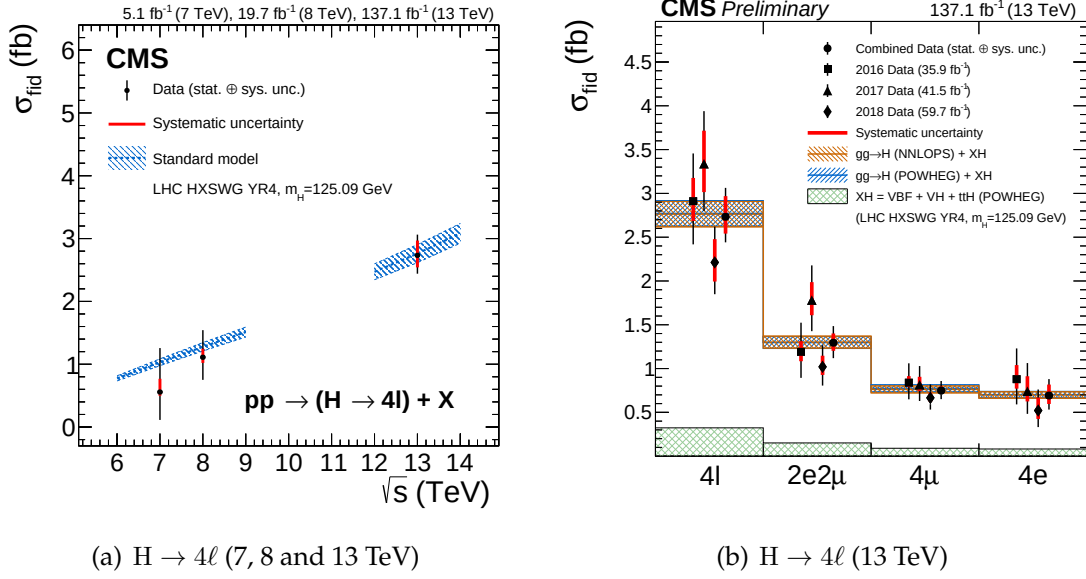


FIGURE 1.14: The measured fiducial cross section as a function of $\sqrt{s}$ (left). The measured inclusive fiducial cross section in different final states (right). The acceptance is calculated using at $\sqrt{s}=13$ and HRES [51, 52] at $\sqrt{s}=7$ and 8 and the total gluon fusion cross section and uncertainty are taken from Ref. [53]. The fiducial volume for $\sqrt{s}=6-9$ uses the lepton isolation definition from Ref. [54], while for $\sqrt{s}=12-14$ the definition described in the text is used.

TABLE 1.13: Measured $H \rightarrow 4\ell$ Inclusive fiducial cross section in $m_{4\ell}$ ranging from 105 GeV to 140 GeV in proton collisions at 13 TeV.

Fiducial cross section $H \rightarrow 4\ell$ at 13 TeV	
Measured	$2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\text{stat.})^{+0.24}_{-0.19}(\text{syst.})$ fb (model)
$gg \rightarrow H(\text{POWHEG}) + XH$	$2.77^{+0.51}_{-0.39}$ fb
$gg \rightarrow H(\text{NNLOPS}) + XH$	$2.76^{+0.14}_{-0.14}$ fb

Measured integrated $H \rightarrow 4\ell$ fiducial cross section using 13 TeV data is in nice agreement with theoretical predictions within corresponding uncertainties. Dominant and sub-dominant uncertainties on the measurement are systematic and statistical uncertainties which are about 24% and 23% respectively. The theoretical uncertainty is about 11%.

TABLE 1.14: Measured $H \rightarrow 4\ell$ integrated or inclusive fiducial cross section performed in $m_{4\ell}$ ranging from 105 GeV to 140 GeV.

Fiducial cross section (Inclusive) of $H \rightarrow 4\ell$ (fb) at 13 TeV.				
	$4l$	$2e2\mu$	4μ	$4e$
Measured (stat. $\oplus$ sys.)	$2.73^{+0.23+0.24}_{-0.22-0.19}$	$1.30^{+0.16+0.10}_{-0.14-0.09}$	$0.75^{+0.10+0.04}_{-0.09-0.04}$	$0.69^{+0.14+0.13}_{-0.13-0.10}$
$gg \rightarrow H(\text{POHWHEG}) + XH$	$2.77^{+0.51}_{-0.39}$	$1.30^{+0.xx}_{-0.xx}$	$0.78^{+0.xx}_{-0.xx}$	$0.70^{+0.xx}_{-0.xx}$
$gg \rightarrow H(\text{NNLOPS}) + XH$	$2.76^{+0.14}_{-0.14}$	$1.30^{+0.07}_{-0.07}$	$0.76^{+0.04}_{-0.04}$	$0.70^{+0.04}_{-0.04}$

1.4.2 Measurement of Differential Fiducial Cross Sections

Observed data along with prediction of SM Higgs boson ($m_H = 125.09$) and expected background in each differential bin for chosen observables are shown in Figures 1.15 for production related observables. The differential results are only produced using 13 TeV data. The results of Higgs differential cross section measurements are produced for four-leptons transverse momentum and rapidity, transverse momentum of leading jet, jet multiplicity and are shown in the Figures 1.16. The uncertainty in the theoretical prediction for dominant process i.e. $gg \rightarrow H$ is estimated in the each differential bin of studied observable with POWHEG+JHUGEN which carries specific signal description. The uncertainty in the theoretical prediction for associated production mechanisms are assumed to be constant among differential bins and are obtained from Ref. [55]. The differential cross sections as a function of Higgs transverse momentum and rapidity are sensitive to pQCD calculations of $gg \rightarrow H$ production mode which is mediated by a loop, in which Higgs $p_T(H)$ is anticipated to be balanced with emission of the soft quarks and gluons. The differential distribution as a function rapidity of Higgs also probes PDFs of the colliding protons and representation of the gluon fusion production modes. The differential observables, transverse momentum p_T) of leading jet and jet multiplicity $N(\text{jets})$ probe hard quark and gluon radiations theoretical modelling in the process. Comparison of measurements with theoretical predictions from POWHEG and NNLOPS for the $gg \rightarrow H$ process (displayed in the blue and brown colors, respectively) together with predictions for $XH = \text{VBF}(\text{POWHEG}) + \text{VH} + \text{ttH}(\text{PYTHIA})$ (shown in green) is also shown. The error bars represents statistical and systematic uncertainty. The results are found to be compatible with SM based theoretical predictions within the uncertainties.

While extracting the differential measurements, the fractions of the $4e$, 4μ and

723 $2e2\mu$ contributions to the fiducial cross sections in each bin is allowed to float.
 724 Results where the fractions are set to their SM predictions are included in Ap-
 725 pendix ???. Similarly, results where the model dependence uncertainty is assessed
 726 by taking into account the experimental constraints from the existing CMS mea-
 surements are included in Appendix ??.

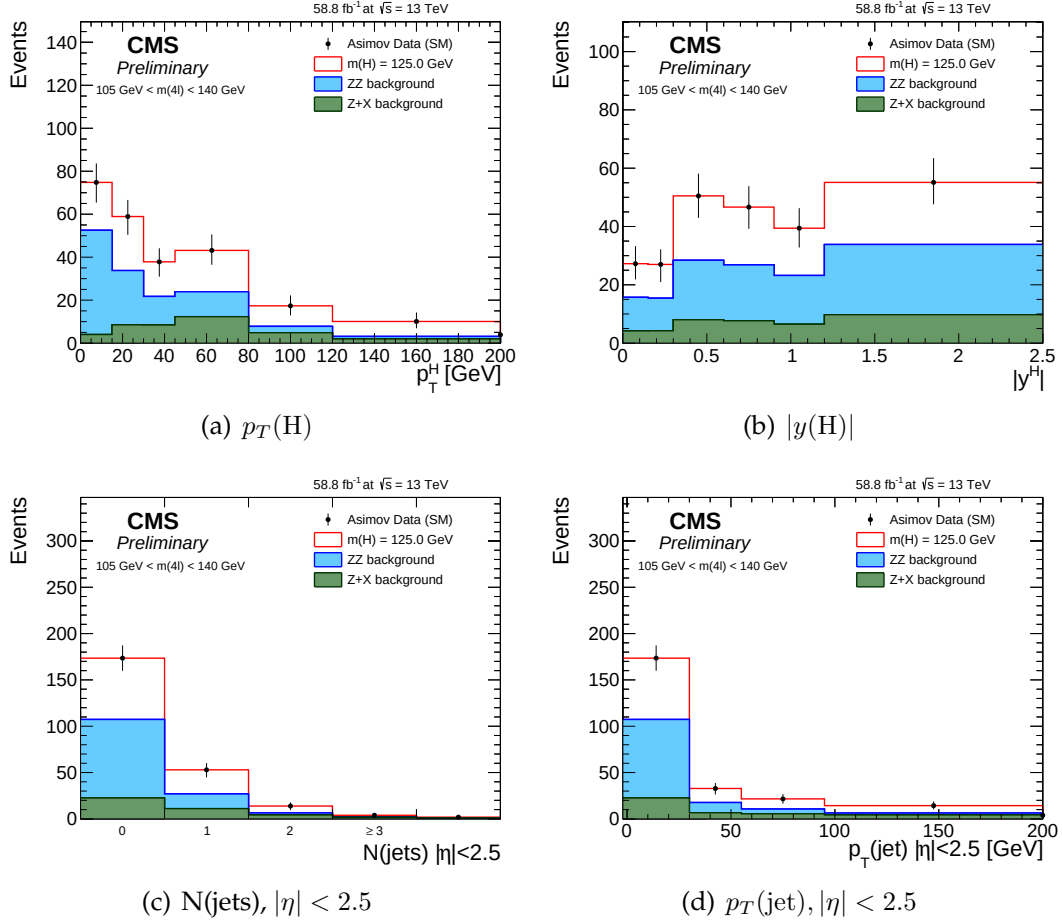


FIGURE 1.15: Expected yield using 2018 Asimov data along with SM signal ($m_H = 125.09$) and background expectations for different observables and for all final states combined, in mass window $105 < m_{4\ell} < 140$ which is used for $H \rightarrow 4\ell$ measurements. The extraction of the signal is performed by exploiting the differences in the expected $m_{4\ell}$ spectra of the signal and the background contributions in each differential bin of an observable.

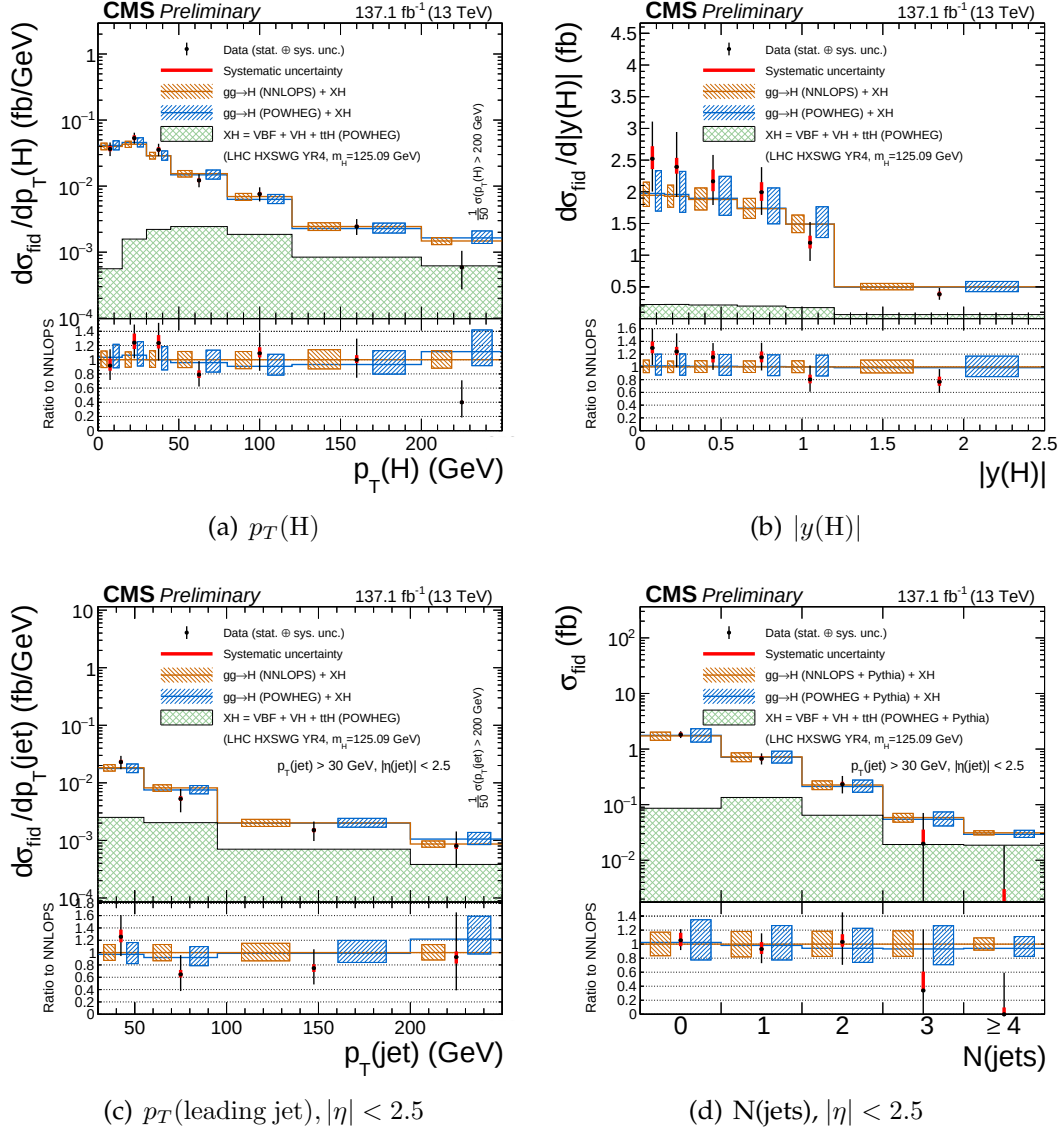


FIGURE 1.16: Differential fiducial cross section results for $p_T(H)$ (top left), $|y(H)|$ (top right), p_T of leading jet (bottom left) and $N(\text{jets})$ (bottom right) in $H \rightarrow 4\ell$ using full Run 2 CMS data and comparison with the theoretical predictions. Systematic uncertainty is indicated by red bars and black bars show total statistical and the systematic uncertainties, combined in quadrature. The blue and brown colors show the theoretical predictions. The acceptances and the theoretical uncertainties in differential bins are computed using POWHEG. Sub-dominant production contributions $XH = \text{VBF} + \text{VH} + t\bar{t}H$ are shown in green separately. The inclusive cross section is normalized to SM estimates calculated at NNLL+NNLO level of accuracy for all theoretical predictions. The systematic uncertainties correspond to the generators accuracy are also taken in to account for differential predictions. The fraction of 4μ , $4e$ and $2e2\mu$ in each differential bin is allowed to float in fit.

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